

Multiple Choice (20 marks)

1. Find $\frac{1}{6} + \frac{1}{8}$.
 - (a) $\frac{7}{24}$
 - (b) $\frac{2}{14}$
 - (c) $\frac{1}{7}$
 - (d) $\frac{1}{4}$
2. The mean of 7 numbers is 15. When an 8th number is added, the mean decreases to 12. What is the 8th number?
 - (a) -12
 - (b) -9
 - (c) 0
 - (d) 8
3. Suppose you toss a fair die three times and it comes up an even number each time. Which of the following is a true statement?
 - (a) By the law of large numbers, the next toss is more likely to be an odd number than another even number.
 - (b) Based on the properties of conditional probability the next toss is more likely to be an even number given that three in a row have been even.
 - (c) Dice actually do have memories, and thus the number that comes up on the next toss will be influenced by the previous tosses.
 - (d) The probability that the next toss will again be even is 0.5.
4. Which is the best buy?
 - (a) 3 bottles of juice for \$6.72
 - (b) 11 bottles of juice for \$24.86
 - (c) 9 bottles of juice for \$20.25
 - (d) 16 bottles of juice for \$35.68
5. Simplify $\frac{2+2i}{-3+4i}$. Express your answer as a complex number in the form $a+bi$, where a and b are real numbers.
 - (a) $2-14i$
 - (b) $\frac{2}{25} + \frac{-14}{25}i$

- (c) $1+0i$
 - (d) $0-1i$
6. The Mills Library has 1,007,199 books. The Springvale Library has 907,082 books. Which of the following is the best estimate of how many more books the Mills...
- (a) 100,000 books
 - (b) 80,000 books
 - (c) 10,000 books
 - (d) 8,000 books
7. Compute $52 + 6 * 2$.
- (a) 116
 - (b) 64
 - (c) 62
 - (d) 118
8. What is the greatest common factor of 252 and 96?
- (a) 6
 - (b) 24
 - (c) 5
 - (d) 12
9. Each story of a building is 14.3 feet tall. If the height of the building is 185.9 feet, how many floors does the building have?
- (a) 10
 - (b) 13
 - (c) 17
 - (d) 9
10. Find the opposite of the number -7.
- (a) 7
 - (b) -7
 - (c) 0.7
 - (d) -7.7

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which method can be used to solve 11×13 ?' is 'Multiply 11×10 and 10×3 , then add the two products.'
2. True or False: The correct answer to 'If 3 cans of pears cost \$2.37 how many cans of pears can you buy for \$9.48?' is '12 cans'.
3. True or False: The correct answer to 'Jennifer is competing in a marathon, which is a 26.2-mile race. If Jennifer runs the first half of the race at 8 mph and the second half...' is '3.74'.
4. True or False: The correct answer to 'Joe and Mike both ran the same race. Joe finished the race 4 minutes before Mike. If Mike finished the race at 4:02 p.m., what time did...' is '3:58 p.m.'.
5. True or False: The correct answer to 'Find the remainder when 25^{1059} is divided by 23.' is '8'.
6. True or False: The correct answer to 'Carlos Montado was born on Saturday, November 9, 2002. On what day of the week will Carlos be 706 days old? (Spell out the entire day of...' is 'Tuesday'.
7. True or False: The correct answer to 'A person walked 3 miles to the east, then turned north and walked 10 miles, then turned west and walked 6 miles, and finally turned south...' is '9.2'.
8. True or False: The correct answer to 'Yesterday, Micah ate an even number of strawberries. He also ate an odd number of apples. Which group could represent the strawberries,...' is 'SSSSS; AAAA'.
9. True or False: The correct answer to 'Two classes take the same exam. Suppose a certain score is at the 40th percentile for the first class and at the 80th percentile for the...' is 'Students in the first class generally scored higher than students in the second class.'.
10. True or False: The correct answer to 'All the students in the sixth grade either purchased their lunch or brought their lunch from home on Monday. 24% of the students...' is '166'.

End of Paper